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RESULTS

Results

01 · ANCOVA

group means adjusted for a covariate

Table 1. Adjusted (estimated marginal) means

Group	Adj. M	SE	95% CI
Blended	65.37	0.71	[63.97, 66.78]
Flipped	70.95	0.71	[69.54, 72.35]
Lecture	60.38	0.71	[58.98, 61.78]

Table 2. ANCOVA

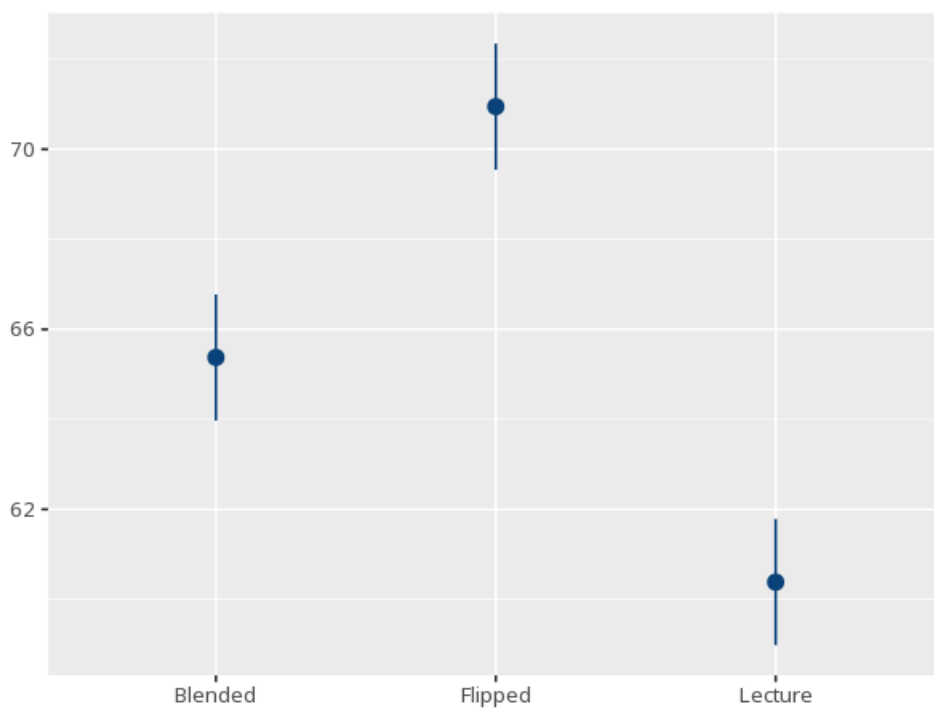
Source	SS	df	MS	F	p	partial η^2 [95% CI]
pretest_score	2086.72	1	2086.72	51.49	<.001	0.18 [0.11, 1.00]
teaching_method	4463.20	2	2231.60	55.07	<.001	0.32 [0.24, 1.00]

assumption checks: homogeneity of regression slopes (factor×covariate interaction), Levene's, & residual normality (Shapiro-Wilk); post-hoc on adjusted means. (slopes $p(\text{pretest_score} \times \text{teaching_method}) = .733$ · Levene $F = 1.44$, $p = .239$ · Shapiro $W = 0.97$, $p < .001$) - the regression slopes look homogeneous across groups - equal variances look reasonable - residual normality looks doubtful; interpret the F-tests with extra caution

Table 3. Post-hoc comparisons (adjusted means)

Pair	M_{diff} (adj.)	SE	p_{adj}	95% CI
Blended - Flipped	-5.57	1.01	<.001	[-7.95, -3.20]
Blended - Lecture	4.99	1.01	<.001	[2.62, 7.37]
Flipped - Lecture	10.57	1.01	<.001	[8.19, 12.94]

Figure. Adjusted means



Understanding the numbers

Source. Names which term the ANCOVA row tests - a covariate or the group factor. Here, the group factor's row is teaching_method.

SS. The variability attributable to that term, after the covariate is accounted for. Here, teaching_method has SS = 4463.20.

df. The degrees of freedom for that term. Here, teaching_method has df = 2.

MS. That term's SS divided by its df. Here, teaching_method has MS = 2231.60.

F. The ratio of that term's mean square to the residual mean square, after removing the covariate's influence. Here, F for teaching_method = 55.07.

p. The probability of an F this large for the group factor if, after adjusting for the covariate, the groups truly had equal means. Here, $p < .001$ for teaching_method - below the $\alpha = 0.05$ threshold.

partial η^2 . The share of variance the group factor explains after removing the covariate's influence. Here, partial η^2 for teaching_method = .32 [.24, 1.00].

Adj. M. The group's estimated mean after statistically removing the covariate's influence. Here, each of the 3 groups has its own adjusted mean in the adjusted-means table above.

SE. The standard error of that adjusted mean, or of a pairwise adjusted-mean difference. Here, each adjusted mean and each pairwise contrast above carries its own SE.

95% CI. The range likely to contain the true adjusted mean, or true adjusted mean difference, for that row. Here, each adjusted mean and each pairwise contrast above has its own 95% CI shown in the tables.

M diff (adj.). The difference between two groups' adjusted means in the post-hoc table. Here, each post-hoc row above reports its own adjusted mean difference for a specific pair.

p (adj). The pairwise p-value for that adjusted-mean comparison, corrected for the number of comparisons made. Here, each pairwise contrast above carries its own multiple-comparison-adjusted p.

How to read this test

Compares group means after statistically removing the influence of a numeric covariate. First confirm the factor×covariate interaction is non-significant (homogeneity of slopes) - if it is significant, the adjusted-means interpretation is not valid and a different model is needed. Then read the Factor row's F/p for the adjusted group effect, and the adjusted means for the covariate-controlled group values. Your significance threshold (α) is 0.05.

APA template: Controlling for the covariate, an ANCOVA gave $F(2,236)=55.07$, $p < .001$, partial $\eta^2=.32$ [0.24, 1.00].

Statistical basis: ANCOVA (group means adjusted for a covariate) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Type III sums of squares (car::Anova) (Fox J, Weisberg S (2019). An R Companion to Applied Regression, Third edition. Sage, Thousand Oaks CA.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.); Partial η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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